

TodoX V8 Released Documentation

Automation Foundation - Database - API - Telegram Render Workflow

Cap nhat: 2026-06-28 | Bao gom muc 12.1 (API Architecture) + muc 22 (Telegram Workflow)

Muc luc:

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TodoX V8 Released Documentation - Automation Foundation

TodoX V8 Released Documentation

Automation Foundation - Database Contract - API Contract - Service Runtime - Deployment Notes

Release date: 2026-06-28 06:47

Item	**Current Release State**
Foundation	PASS - Doctor V3.6 completed with PASS=23, WARN=0, FAIL=0
Database Migration	V1.3 FIX5 used to relax Telegram-specific constraints and support generic API/n8n/manual jobs
Database Contract	V1.0 FIX1 installed in package1-database-contract and used as default generator
Automation API	V1.1 installed as TodoXAutomationAPI on port 8787
Service Drift	TodoXAutomationAPI AppDirectory standardized to E:\N8N.ANHDUC\todoX\automation-api
Next Focus	Build n8n workflow: Telegram /render link -> create job -> TikTok ingest -> scene analysis -> remake script -> JSON artifacts -> MinIO link

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1. Executive Summary

TodoX V8 is the stabilized foundation layer for the TodoX automation/render system. The key architectural decision is to move business state out of n8n and into a dedicated Automation API backed by PostgreSQL and MinIO. n8n will become a workflow orchestration layer instead of the place where all business logic, state, file structure, and retry rules live.

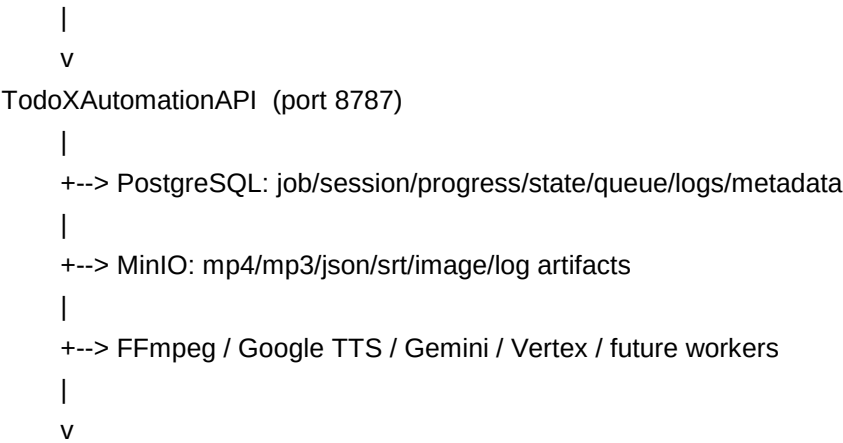
At this milestone, TodoX Foundation has a working service, database schema, database contract, API contract, MinIO storage convention, and Doctor validation process. The system is now ready to shift attention to the actual workflow result: Telegram /render link -> TikTok video processing -> scene analysis -> script remake -> artifacts -> Telegram response.

2. What We Completed

Area	**Result**	**Notes**
TodoX Root	Standardized	All main system paths now converge under E:\N8N.ANHDUC\todoX.
Render API Drift	Fixed	Service no longer runs from old minio_iis_setup path. New runtime is TodoXAutomationAPI.
PostgreSQL Foundation	PASS	Database tables for jobs, sessions, steps, scenes, artifacts, logs, queue, workers are present.
Database Migration	PASS after V1.3 FIX5	Handled legacy schema drift: id text default, chat_id nullable, optional Telegram fields.
Database Contract	V1.0 FIX1	Updated generator handles default expressions containing pipe characters and writes contract JSON.
Doctor	V3.6 RELEASE PASS	Doctor confirmed contracts, service, database tables, and API /health.
Automation API	V1.1 installed	Service is running on port 8787 and Create Job passed after schema fix.
MinIO	Standardized	Bucket todox-render and alias todox-minio are used.
JSON-first	Adopted	Scene scripts, caption, analysis, metadata, and progress should be JSON first.

3. TodoX V8 Standard Architecture

Telegram / n8n / Dashboard



n8n workflow orchestration and Telegram response

Important rule: n8n should not be the long-term system of record. PostgreSQL is the system of record for state and MinIO is the system of record for files/artifacts.

4. Server Paths and Runtime Configuration

Key	**Value**
TodoX root	E:\N8N.ANHDUC\todoX
Automation API path	E:\N8N.ANHDUC\todoX\automation-api
Legacy render-api path	E:\N8N.ANHDUC\todoX\render-api
Work path	E:\N8N.ANHDUC\todoX\work
Logs path	E:\N8N.ANHDUC\todoX\logs
Temp path	E:\N8N.ANHDUC\todoX\temp
Config path	E:\N8N.ANHDUC\todoX\config
Docs path	E:\N8N.ANHDUC\todoX\docs
Database path	E:\N8N.ANHDUC\todoX\database
Database backups	E:\N8N.ANHDUC\todoX\database\backups
Contracts path	E:\N8N.ANHDUC\todoX\contracts
Database contract path	E:\N8N.ANHDUC\todoX\contracts\database
FFmpeg path	E:\N8N.ANHDUC\todoX\ffmpeg\bin\ffmpeg.exe
Vertex SA key	E:\N8N.ANHDUC\todoX\keys\todox-vertex-sa.json
NSSM path	C:\ProgramData\chocolatey\bin\nssm.exe
MinIO client path	C:\minio\tools\mc.exe

5. Secrets and Environment Variables

The following environment values were confirmed during TodoX V8 Foundation work and should be stored in TodoX V8 documentation and in the runtime service configuration. Because this document is intended as a deployment record, sensitive values are included as requested.

Variable	**Value**
TODOX_RENDER_API_SECRET	todox-render-2026
TODOX_RENDER_API_PORT	8787
TODOX_AUTOMATION_API_PORT	8787
TODOX_ROOT	E:\N8N.ANHDUC\todoX
TODOX_WORK_DIR	E:\N8N.ANHDUC\todoX\work
TODOX_LOG_DIR	E:\N8N.ANHDUC\todoX\logs
TODOX_TEMP_DIR	E:\N8N.ANHDUC\todoX\temp
TODOX_CONFIG_DIR	E:\N8N.ANHDUC\todoX\config
TODOX_FFMPEG_PATH	E:\N8N.ANHDUC\todoX\ffmpeg\bin\ffmpeg.exe
FFMPEG_PATH	E:\N8N.ANHDUC\todoX\ffmpeg\bin\ffmpeg.exe
GOOGLE_APPLICATION_CREDENTIALS	E:\N8N.ANHDUC\todoX\keys\todox-vertex-sa.json
TODOX_GOOGLE_APPLICATION_CREDENTIALS	E:\N8N.ANHDUC\todoX\keys\todox-vertex-sa.json
TODOX_PGHOST	127.0.0.1
TODOX_PGPORT	5432

TODOX_PGDATABASE	todox
TODOX_PGUSER	postgres
TODOX_PGPASSWORD	Q3sNSrRwxquzAhO24pjfHovL
TODOX_MINIO_BUCKET	todox-render
TODOX_MINIO_CONSOLE_BASE_URL	https://minio-console.tuvangiaiphapta.vn/browser
TODOX_MC_PATH	C:\minio\tools\mc.exe
TODOX_MC_CONFIG_DIR	E:\N8N.ANHDUC\todoX\minio-config
TODOX_MC_ALIAS	todox-minio
TODOX_MINIO_UPLOAD_ENABLED	true
TODOX_TTS_ENABLED	true
TODOX_TTS_LANGUAGE	vi-VN
TODOX_TTS_VOICE	vi-VN-Neural2-A
TODOX_TTS_FALLBACK_ON_ERROR	true
TODOX_VEO_ENABLED	false
TODOX_SCENE_DURATION_SEC	8

5.1 NSSM command to inspect service secrets

C:\ProgramData\chocolatey\bin\nssm.exe get TodoXAutomationAPI AppEnvironmentExtra

C:\ProgramData\chocolatey\bin\nssm.exe get TodoXRenderAPI AppEnvironmentExtra

6. Windows Services and Port Map

Service / Component	**Port**	**Status / Purpose**
TodoXAutomationAPI	8787	Primary TodoX Automation API service. Replaces old Render API as the main runtime service.
TodoXRenderAPI	8787 legacy	Legacy compatible service name. Should not conflict with TodoXAutomationAPI. Use only for old workflows if needed.
TodoXTikTokIngestAPI	8080	Legacy TikTok/ingest/FFmpeg-related API service.
n8n	8997	Workflow orchestration UI and webhook runtime.
MinIO S3	8900	Object storage API.
MinIO Console	8901	Storage console. Public console base uses https://minio-console.tuvangiaiphapta.vn/browser.
PostgreSQL 17	5432	Database todox.
Redis	6379	Reserved for future queue/worker scaling; may be absent now.

6.1 TodoXAutomationAPI NSSM standard

Key	**Value**
Service name	TodoXAutomationAPI
Display name	TodoX Automation API

Application	C:\Program Files\nodejs\node.exe
AppDirectory	E:\N8N.ANHDUC\todoX\automation-api
AppParameters	app.js
Start type	SERVICE_AUTO_START

7. PostgreSQL Database Foundation

PostgreSQL database todox is the state foundation. It stores job lifecycle, Telegram/API sessions, steps, scenes, artifact metadata, queue items, worker status, config, and logs. MinIO stores heavy files and JSON artifacts; PostgreSQL stores pointers and operational metadata.

Key	**Value**
Host	127.0.0.1
Port	5432
Database	todox
User	postgres
Password	Q3sNSrRwxquzAhO24pjfHovL
psql path	C:\Program Files\PostgreSQL\17\bin\psql.exe
pg_dump path	C:\Program Files\PostgreSQL\17\bin\pg_dump.exe

8. Database Table Details

The following table structures describe the TodoX Foundation contract. The actual database has legacy columns in some tables; the contract must allow generic jobs from Telegram, n8n, API, or Dashboard.

todox_jobs

Column	**Type**	**Rule / Default**	**Purpose**
id	text	NOT NULL, default job_ + md5(random/clock_timestamp)	Legacy primary internal row id; V1.3 FIX4/FIX5 added text default.
job_id	varchar	nullable but should be populated	Public job identifier used by API/n8n/MinIO.
chat_id	text	nullable	Telegram-specific; must not be required for API/manual jobs.
source	varchar	default api / legacy	Source: telegram, n8n, api, manual, doctor.
source_url	text	nullable	Original TikTok/FB/YT/source link.
command	text	nullable	Command such as /render.
status	text/varchar	default created/PLANNED depending legacy	Job lifecycle status.
progress_percent	integer	default 0	Progress 0-100.
minio_prefix	text	nullable	jobs/{jobId}/ prefix.
options	jsonb	default {}	Job options: language, jsonFirst, render flags.
created_at	timestampz	default now()	Created time.

updated_at	timestampz	default now()	Updated time.
completed_at	timestampz	nullable	Completion time.
error	jsonb/text	nullable	Error details.

todox_sessions

Column	**Type**	**Rule / Default**	**Purpose**
id	bigserial/int	primary key	Internal row id.
session_id	varchar	unique	Session identifier.
job_id	varchar	nullable	Related job.
channel	varchar	default api	telegram, api, n8n, dashboard.
user_ref	text	nullable	User reference.
chat_id	text	nullable	Telegram chat id when used.
status	varchar	default active	Session status.
context	jsonb	default {}	Session context.
created_at	timestampz	default now()	Created time.
updated_at	timestampz	default now()	Updated time.

todox_job_steps

Column	**Type**	**Rule / Default**	**Purpose**
id	bigserial	primary key	Internal row id.
job_id	varchar	not null	Related job id.
step_key	varchar	not null	Stable key: source_ingest, scene_analyze, tts, render.
step_name	text	nullable	Human-readable step name.
status	varchar	default pending	pending, running, completed, failed.
attempt	integer	default 0	Retry attempt.
started_at	timestampz	nullable	Start time.
finished_at	timestampz	nullable	Finish time.
input	jsonb	nullable	Step input.
output	jsonb	nullable	Step output.
error	jsonb	nullable	Step error.

todox_scenes

Column	**Type**	**Rule / Default**	**Purpose**
id	bigserial/int/text legacy	primary/internal	Internal row id.
job_id	varchar	not null/legacy nullable	Related job id.
scene_index	integer	default 1	Scene number starting at 1.

version	integer	default 1	Scene version for rerender.
status	varchar	default created	Scene status.
start_sec	numeric	nullable	Scene start.
end_sec	numeric	nullable	Scene end.
summary	text	nullable	Scene summary.
characters	jsonb	default []	Characters/persons in scene.
analysis	jsonb	default {}	Gemini analysis result.
remake_script	jsonb	default {}	New script payload.
created_at	timestampz	default now()	Created time.
updated_at	timestampz	default now()	Updated time.

todox_artifacts

Column	**Type**	**Rule / Default**	**Purpose**
id	bigserial	primary key	Internal row id.
job_id	varchar	not null	Related job id.
scene_index	integer	nullable	Scene if artifact is scene-specific.
version	integer	default 1	Artifact version.
artifact_type	varchar	not null	metadata, source_video, scene_analysis, remake_script, voice, caption_json, scene_video, final_video, log, image.
format	varchar	not null	json, mp4, mp3, wav, srt, vtt, png, jpg, log.
storage_provider	varchar	default minio	Storage backend.
bucket	varchar	nullable	MinIO bucket.
object_key	text	not null	MinIO object key.
public_url	text	nullable	Public/console URL.
console_url	text	nullable	Console URL.
size_bytes	bigint	nullable	File size.
checksum	text	nullable	SHA256 or similar.
metadata	jsonb	default {}	Additional metadata.
created_at	timestampz	default now()	Created time.

todox_job_logs

Column	**Type**	**Rule / Default**	**Purpose**
id	bigserial	primary key	Internal row id.
job_id	varchar	nullable	Related job id.
level	varchar	default INFO	INFO, WARN, ERROR.
message	text	not null	Log message.

data	jsonb	nullable	Structured details.
created_at	timestampz	default now()	Created time.

todox_config

Column	**Type**	**Rule / Default**	**Purpose**
config_key	varchar	primary key	Config key.
config_value	jsonb	not null	Config value.
description	text	nullable	Description.
updated_at	timestampz	default now()	Updated time.

todox_queue

Column	**Type**	**Rule / Default**	**Purpose**
id	bigserial	primary key	Internal queue id.
queue_name	varchar	not null	default, tts, render, finalize, upload.
job_id	varchar	nullable	Related job id.
task_type	varchar	not null	Task type.
payload	jsonb	default {}	Task payload.
status	varchar	default queued	queued, locked, done, failed.
priority	integer	default 100	Lower number can mean higher priority.
attempt	integer	default 0	Attempt count.
available_at	timestampz	default now()	When task can run.
locked_at	timestampz	nullable	Lock time.
locked_by	varchar	nullable	Worker id.
created_at	timestampz	default now()	Created time.
updated_at	timestampz	default now()	Updated time.

todox_workers

Column	**Type**	**Rule / Default**	**Purpose**
worker_id	varchar	primary key	Worker identifier.
worker_type	varchar	nullable	gemini, tts, ffmpeg, upload, publish.
status	varchar	default unknown	online, offline, busy, unknown.
last_heartbeat_at	timestampz	nullable	Last heartbeat.
metadata	jsonb	default {}	Worker details.

todox_scene_render_tasks

Column	**Type**	**Rule / Default**	**Purpose**
id	bigserial	primary key	Internal row id.

job_id	varchar	not null	Related job.
scene_index	integer	not null	Scene index.
version	integer	default 1	Scene version.
status	varchar	default queued	Task status.
payload	jsonb	default {}	Render payload.
result	jsonb	nullable	Render result.
error	jsonb	nullable	Render error.
created_at	timestamptz	default now()	Created time.
updated_at	timestamptz	default now()	Updated time.

todox_foundation_migrations

Column	**Type**	**Rule / Default**	**Purpose**
migration_key	varchar	primary key	Migration key such as database_v1_3_fix5.
migration_name	text	not null	Human-readable migration name.
applied_at	timestamptz	default now()	Applied time.
notes	jsonb	default {}	Migration notes.

9. Database Migration History

Migration	**Purpose**	**Final Status**
Foundation Migration V1.0	Initial schema proposal. Failed due to existing legacy schema drift.	Superseded
V1.1	Fixed PowerShell \$\$ escaping and legacy schema version issue.	Superseded
V1.2	Avoided schema_version conflict and created foundation tables.	PASS
V1.3	Tried to relax Telegram-specific NOT NULL constraints.	Superseded
V1.3 FIX1/FIX2/FIX3/FIX4	Resolved psql invocation, BOM, id default, text id issues.	Superseded
V1.3 FIX5	Final robust schema relaxation; only alters optional columns when they exist.	Current final for this stage

9.1 Key database lessons

The existing todox_jobs.id column is text/varchar, not numeric. Future scripts must not assume BIGSERIAL.

Telegram fields like chat_id belong in session context and must not block generic jobs.

Migration scripts must check whether optional columns exist before ALTER COLUMN.

SQL files should be written without UTF-8 BOM when executed by psql on Windows.

Database Contract must be regenerated after every schema migration.

10. Database Contract V1.0 FIX1

Database Contract captures the real PostgreSQL schema into JSON files. It is the bridge between PostgreSQL and the Automation API so code does not assume a schema that differs from the running database.

Key	**Value**
Contract root	E:\N8N.ANHDUC\todoX\contracts\database
Main contract file	E:\N8N.ANHDUC\todoX\contracts\database\database-contract.json
Per-table files	E:\N8N.ANHDUC\todoX\contracts\database\tables*.table.json
Generator folder	E:\N8N.ANHDUC\todoX\automation-api\todox_foundation_v36_api_v11\package1-database-contract
Generator script	01-Generate-Database-Contract.ps1 - replaced with FIX1 version

```
cd E:\N8N.ANHDUC\todoX\automation-api\todox_foundation_v36_api_v11\package1-database-contract
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass -Force
.\01-Generate-Database-Contract.ps1
```

11. Automation API Contract V1.0

Automation API Contract defines REST endpoint names, request/response structure, error format, JSON schemas, and storage naming conventions. It is installed under TodoX contracts and is used to build API, n8n workflows, and future .NET Dashboard clients.

Key	**Value**
Contract file	E:\N8N.ANHDUC\todoX\contracts\automation-contract.json
OpenAPI file	E:\N8N.ANHDUC\todoX\contracts\openapi.yaml
JSON schemas	E:\N8N.ANHDUC\todoX\contracts\schemas*.json
Swagger URL	http://127.0.0.1:8787/docs
Auth header	X-TodoX-Secret
Secret	todox-render-2026

12. TodoX Automation API V1.1

TodoX Automation API V1.1 is the runtime service that replaces the old render-only thinking. It inserts jobs into PostgreSQL, writes JSON-first artifacts, uploads artifacts to MinIO, and exposes queue/worker endpoints for future render workers.

Endpoint	**Purpose**	**Current Foundation Status**
GET /health	Check service, version, DB connectivity	PASS
POST /api/jobs	Create automation job	PASS after Database V1.3 FIX5
GET /api/jobs/{jobId}	Get job detail	Available
GET /api/jobs/{jobId}/status	Get job status/progress	Available
GET /api/jobs/{jobId}/artifacts	List artifacts	Available
POST /api/storage/upload-json	Write JSON artifact locally and MinIO	Available
POST /api/storage/upload-file	Upload binary file artifact	Available in V1 implementation
POST /api/queue/enqueue	Add queue item	Available

POST /api/queue/next	Worker pulls next task	Available
POST /api/workers/heartbeat	Worker heartbeat	Available
POST /api/source/tiktok/ingest	Register or ingest TikTok source	Foundation endpoint
POST /api/scenes/analyze	Store scene analysis	Foundation endpoint
POST /api/scenes/remake-script	Store remake script	Foundation endpoint
POST /api/tts/google	Queue/register Google TTS request	Foundation endpoint
POST /api/caption/create	Create caption JSON/SRT data	Foundation endpoint
POST /api/video/render-scene	Queue scene render task	Foundation endpoint
POST /api/video/finalize	Queue final video task	Foundation endpoint

12.1 API Internal Architecture (Bo sung tu Diagnostic 2026-06-28)

Phan nay ghi lai cau truc code thuc te cua TodoXAutomationAPI, xac nhan qua diagnostic log, de cac lan nang cap sau khong lam hong service dang chay.

File dang chay that su

NSSM service TodoXAutomationAPI chay file:

- Application: C:\Program Files\nodejs\node.exe
- AppDirectory: E:\N8N.ANHDUC\todoX\automation-api
- AppParameters: app.js
- Entry that su: E:\N8N.ANHDUC\todoX\automation-api\app.js (9155 bytes)

Luu y: ton tai cac ban app.js khac (todox_automation_api_v1, package3-automation-api-v11) nhung KHONG phai ban dang chay. Chi sua app.js o root automation-api.

Tech stack

- Framework: Express 4.18
- DB driver: pg 8.13 (node-postgres) voi connection Pool
- Upload: multer
- Docs: swagger-ui-express doc tu contracts/openapi.yaml
- Auth: middleware requireSecret kiem tra header x-todox-secret

Cau truc thu muc

Path	Vai tro
automation-api/app.js	Entry point, dang ky tat ca route
automation-api/src/lib/db.js	Export: pool, query, getColumns, getColumnMap, hasDefault, numericType
automation-api/src/lib/storage.js	Export: ensureDir, writeJson, key, url (ghi JSON + upload MinIO qua mc.exe)
automation-api/src/middleware/auth.js	Export: requireSecret
automation-api/src/routes/	Thu muc route (hien TRONG - dung de extend)
automation-api/src/controllers/	Thu muc controller (hien trong)

Pattern chuan trong app.js

Response helper:

- ok(data, traceId) -> {ok:true, data, error:null, traceId, time}

- fail(code, message, detail, traceId) -> {ok:false, data:null, error:{...}, traceId, time}

DB helper:

- insertDynamic(table, data): chỉ insert cột tồn tại trong bảng (dùng getColumnMap), an toàn với schema drift
- query(sql, params): truy vấn trực tiếp

Auth: app.use('/api', requireSecret) - mọi route /api/* đều qua auth.

Trạng thái endpoint Foundation (QUAN TRỌNG)

Các endpoint sau hiện tại chỉ là STUB - chỉ ghi log và writeJson, CHƯA xử lý thật:

- /api/source/tiktok/ingest
- /api/scenes/analyze
- /api/scenes/remake-script
- /api/tts/google
- /api/caption/create
- /api/video/render-scene
- /api/video/finalize

Chúng được đăng ký bằng vòng lặp route đơn giản (app.js dòng 162-172). Khi cần xử lý thật (vd TTS tạo mp3 thật, concat video thật), phải thay stub bằng implementation thực - nên làm qua file route riêng trong src/routes/ và require vào app.js, KHÔNG sửa trực tiếp vòng lặp stub.

upload-json field name

Endpoint POST /api/storage/upload-json nhận body:

- jobId (bắt buộc)
- relativePath (vd 'analysis/scene_analysis_v1.json') - KHÔNG phải object_key
- data (nội dung JSON)
- artifactType, sceneIndex, version (optional)

Workflow phải dùng relativePath, không dùng object_key.

Quy tắc mở rộng an toàn

1. Không sửa trực tiếp các hàm lỗi của app.js (insertJob, insertDynamic, ok, fail).
2. Endpoint mới -> tạo file trong src/routes/, export 1 hàm nhận (app, deps).
3. Require file route vào app.js NGAY TRƯỚC khi dùng app.use 404 handler.
4. Backup app.js trước mỗi lần sửa.
5. Sau khi sửa: restart service + chạy Doctor V3.6 + regenerate Database Contract.

13. MinIO Storage and Artifact Convention

MinIO stores all generated and source artifacts. PostgreSQL stores metadata and object links. Artifacts must be grouped by job and version so individual scenes can be re-rendered later without overwriting history.

jobs/{jobId}/

video-resource/

metadata/

analysis/

remake/

scene-video/

scene-voice/

scene-caption/

scene-image/

scene-render/
video-final/
logs/
benchmark/

Folder	**Purpose**	**Example**
video-resource	Original downloaded source video or extracted resources	source_video_v1.mp4
metadata	job manifest, system metadata, compact status files	job_manifest_v1.json
analysis	Scene analysis and transcript JSON	scene_analysis_v1.json
remake	New script/remake JSON	scene_001_script_v1.json
scene-video	Rendered scene MP4 files	scene_001_v1.mp4
scene-voice	Voice/TTS outputs	scene_001_voice_v1.mp3
scene-caption	Caption JSON/SRT/VTT	scene_001_caption_v1.json
scene-image	Generated image/keyframe assets	scene_001_image_v1.png
scene-render	Intermediate render payloads/results	scene_001_render_payload_v1.json
video-final	Final assembled output	final_v1.mp4
logs	Job logs/progress	progress_v1.json
benchmark	Timing and debug metrics	timing_v1.json

14. JSON-first Data Standard

TodoX V8 uses JSON as the canonical data format. Text formats such as TXT, SRT, VTT are exports only. n8n, API, worker, and dashboard should read JSON directly whenever possible.

```
{  
  "jobId": "job_20260628_001",  
  "sceneIndex": 1,  
  "version": 1,  
  "hook": "...",  
  "script": "...",  
  "voiceText": "...",  
  "captionText": "...",  
  "prompt": "...",  
  "characters": ["..."],  
  "metadata": {}  
}
```

15. Doctor V3.6 Validation Standard

Doctor V3.6 checks the foundation layer before building workflow. Latest confirmed run returned PASS=23, WARN=0, FAIL=0.

Doctor Check	**Purpose**
Service TodoXAutomationAPI	Confirm service exists and runs.
Service Drift	Confirm AppDirectory is E:\N8N.ANHDUC\todoX\automation-api.

Automation Contract	Confirm automation-contract.json exists.
OpenAPI	Confirm openapi.yaml exists.
Database Contract	Confirm database-contract.json exists.
PostgreSQL Tables	Confirm foundation tables exist.
PostgreSQL Columns	Confirm key todox_jobs columns exist.
API /health	Confirm API is live and reports version.

```
cd E:\N8N.ANHDUC\todoX\automation-api\todox_foundation_v36_api_v11\package2-doctor-v36
.\TodoX-Doctor-V3.6-RELEASE.ps1
```

16. Operating Procedures

16.1 Check PostgreSQL manually

```
$env:PGPASSWORD="Q3sNSrRwxquzAhO24pjfHovL"
& "C:\Program Files\PostgreSQL\17\bin\psql.exe"
-h 127.0.0.1
-p 5432
-U postgres
-d todox
-c "select current_user,current_database();"
```

16.2 Check TodoXAutomationAPI

```
Invoke-RestMethod -Uri "http://127.0.0.1:8787/health"
```

16.3 Test API V1.1

```
cd E:\N8N.ANHDUC\todoX\automation-api\todox_foundation_v36_api_v11\package3-automation-api-v11
.\Test-TodoX-Automation-API-V1.1.ps1
```

17. Standard Upgrade Workflow

Step	**Command / Action**	**Reason**
1	Run migration if schema changes	Update database safely.
2	Run Database Contract generator FIX1	Snapshot real schema.
3	Update/verify Automation API Contract	Keep endpoint contract aligned.
4	Install/restart TodoXAutomationAPI	Run API matching current contract.
5	Run Doctor V3.6	Confirm service, contract, DB, and API health.
6	Run API test script	Confirm Create Job, upload JSON, artifacts.
7	Only then update workflow	Avoid workflow-level bugs caused by foundation drift.

18. n8n Workflow Integration Plan

The next workflow must be built on top of the stable API foundation. The initial goal should not be full final video generation. It should be a reliable Telegram workflow that creates a job, downloads/analyzes a TikTok link, creates a new scene script, uploads JSON artifacts to MinIO, and returns compact Telegram links.

Workflow Step	**Owner**	**Output**
Telegram receives /render <TikTok link>	n8n	sourceUrl and chatId.
POST /api/jobs	n8n -> API	jobId, minioPrefix, job_manifest_v1.json.
Download TikTok MP4	n8n or ingest API	video-resource/source_video_v1.mp4 artifact.
Scene analysis	Gemini via n8n/API	analysis/scene_analysis_v1.json.
Create remake script	Gemini via n8n/API	remake/scene_001_script_v1.json etc.
Upload JSON artifacts	API storage upload-json	MinIO links stored in todox_artifacts.
Telegram compact reply	n8n	Job ID + MinIO links, not long text.
Future: TTS/render scene	Worker/API	scene-voice and scene-video artifacts.

19. Troubleshooting Notes

Symptom	**Cause Found**	**Resolution**
Render API drift keeps appearing	NSSM AppDirectory pointed to old minio_iis_setup path	Standardized to TodoXAutomationAPI at E:\N8N.ANHDUC\todoX\automation-api.
PostgreSQL auth failed for Administrator	Script invocation of psql did not pass user/env correctly	Use direct PowerShell invocation with & and set PGPASSWORD.
SQL error at '»¿BEGIN	UTF-8 BOM in SQL file	Write SQL as ASCII/no BOM.
Create Job failed: id null	todox_jobs.id text NOT NULL without default	V1.3 FIX4/FIX5 set text default job_ + md5.
Create Job failed: chat_id null	Telegram-specific field was NOT NULL in generic jobs table	V1.3 FIX5 made chat_id nullable.
Database Contract failed parsing default	column_default contained pipe character	Database Contract FIX1 base64-encodes default expressions before parsing.

20. Next Actions

Priority	**Next Work**	**Expected Result**
1	Run Database Contract FIX1 after final schema	Fresh database-contract.json reflects FIX5 schema.
2	Run Doctor V3.6 and API V1.1 test	Foundation PASS and API job/artifact flow confirmed.
3	Build n8n Workflow V1: Telegram -> Create Job -> TikTok analysis -> remake JSON -> MinIO links	First real workflow result.
4	Add Google TTS endpoint/worker for first scene voice	scene-voice artifact generated.
5	Add first scene render with captions	scene-video artifact generated.

6	Add queue worker pattern for parallel scene processing	Scale to multi-scene video generation.
---	--	--

Appendix A. PowerShell Commands

Inspect service environment

C:\ProgramData\chocolatey\bin\nssm.exe get TodoXAutomationAPI AppEnvironmentExtra

PostgreSQL direct test

```
$env:PGPASSWORD="Q3sNSrRwxquzAhO24pjfHovL"
& "C:\Program Files\PostgreSQL\17\bin\psql.exe" -h 127.0.0.1 -p 5432 -U postgres -d todox -c "select current_user,current_database();"
```

Generate Database Contract FIX1

```
cd E:\N8N.ANHDUC\todoX\automation-api\todox_foundation_v36_api_v11\package1-database-contract
.\01-Generate-Database-Contract.ps1
```

Doctor V3.6

```
cd E:\N8N.ANHDUC\todoX\automation-api\todox_foundation_v36_api_v11\package2-doctor-v36
.\ToDoX-Doctor-V3.6-RELEASE.ps1
```

API V1.1 test

```
cd E:\N8N.ANHDUC\todoX\automation-api\todox_foundation_v36_api_v11\package3-automation-api-v11
.\Test-TodoX-Automation-API-V1.1.ps1
```

Appendix B. API Endpoint Summary

Method	**Endpoint**	**Group**
GET	/health	system
POST	/api/jobs	jobs
GET	/api/jobs/{jobId}	jobs
GET	/api/jobs/{jobId}/status	jobs
GET	/api/jobs/{jobId}/artifacts	artifacts
POST	/api/storage/upload-json	storage
POST	/api/storage/upload-file	storage
POST	/api/queue/enqueue	queue
POST	/api/queue/next	queue
POST	/api/workers/heartbeat	workers
POST	/api/source/tiktok/ingest	source
POST	/api/scenes/analyze	scenes
POST	/api/scenes/remake-script	scenes
POST	/api/tts/google	tts
POST	/api/caption/create	caption
POST	/api/video/render-scene	video

POST	/api/video/finalize	video
------	---------------------	-------

Appendix C. SQL/Data Contract Notes

The most important database rule after V1.3 FIX5: `todox_jobs` is generic and must support jobs created from Telegram, API, n8n, manual, or dashboard. Therefore Telegram-specific fields must not be mandatory in `todox_jobs`. Telegram state belongs in `todox_sessions` or context JSON.

Recommended job creation minimum fields:

- job_id
- source
- source_url
- command
- status
- progress_percent
- minio_prefix
- options
- created_at
- updated_at

Do not require:

- chat_id
- message_id
- user_id
- username

Confidential deployment document - includes runtime secrets as requested by owner

22. Telegram Render Workflow (V8 TG)

Bo workflow n8n dieu khien qua Telegram, render video song song 3 worker.

22.1 Luong hoat dong

User go: `/render <tiktok_url> [duration]`

1. WF1 Main Orchestrator (Telegram trigger)

- Load `todox_settings` -> `CONFIG_*`
- Parse lenh, validate
- Telegram: "Dang phan tich video"
- POST `/api/jobs` -> jobId
- Download MP4 -> Gemini analyze
- Telegram MOC 1: "Phan tich xong, N scene"
- Gemini remake script
- Telegram MOC 2: "Kich ban san sang, bat dau render"
- Init counter `worker_done=0`, `worker_total=3`
- Fan-out 3 workers song song

2. WF2a/2b/2c Workers (song song)

- Moi worker render nhom scene cua minh: Veo3 -> TTS -> mux caption
- Xong batch: POST `/api/jobs/worker-complete` (tang counter)
- Telegram MOC 3: "Worker X xong (X/3)"

- Worker cuối (3/3) gọi Concat

3. WF3 Concat & Deliver

- Gom tất cả scene theo thứ tự
- FFmpeg concat -> final_v1.mp4 -> MinIO
- Update job completed
- Telegram MOC 4: "VIDEO HOÀN TẤT + link"

22.2 Cách chia scene cho 3 worker

3 là số lượng song song có định, không phải giới hạn scene.

Scenes chia đều: $bSize = \lceil total/3 \rceil$

Tổng scene	Worker 1	Worker 2	Worker 3
3	1	2	3
6	1,2	3,4	5,6
10	1,2,3,4	5,6,7	8,9,10

Mỗi worker render tuần tự các scene trong nhóm. Thời gian ~ nhóm lớn nhất, nhanh gấp ~3 lần số vòng tuần tự.

22.3 Endpoint mới: POST /api/jobs/worker-complete

Để đảm bảo atomic để biết khi nào cả 3 worker xong, đảm bảo Concat chỉ chạy 1 lần.

Request:

POST /api/jobs/worker-complete

Header: X-TodoX-Secret

Body: { jobId, workerId, rendered_scenes[], skipped_scenes[] }

Response:

{ ok, worker_done, worker_total, is_last_worker, all_rendered_scenes[], all_skipped_scenes[] }

Implementation: tạo file src/routes/worker-complete.js, require vào app.js.

Dùng todox_jobs.options cho counter, todox_scene_render_tasks cho scene results.

KHÔNG tạo bảng mới.

22.4 Settings Telegram trong todox_settings

key	category	mô tả
telegram_bot_token	telegram	Bot token từ @BotFather
telegram_allowed_chat_ids	telegram	Optional: giới hạn chat_id

22.5 Files workflow

File	Webhook path
TodoX_V8_TG_1_Main_Orchestrator.json	Telegram trigger
TodoX_V8_TG_2a_Worker1.json	/webhook/worker-scene-1
TodoX_V8_TG_2b_Worker2.json	/webhook/worker-scene-2
TodoX_V8_TG_2c_Worker3.json	/webhook/worker-scene-3
TodoX_V8_TG_3_Concat_Deliver.json	/webhook/concat-deliver

22.6 Install scripts (chay theo thu tu)

Script	Tac dung
00_Install-All.ps1	Master: chay tat ca buoc duoi
01_seed_settings.sql + Update-Settings.ps1	Tao bang todox_settings + seed
02_migration_worker_complete.sql	Tao function counter + index
03_Install-Endpoint.ps1	Cai endpoint worker-complete vao API
04_settings_telegram.sql	Them settings Telegram

Sau khi chay xong scripts -> import 5 workflow trong n8n (buoc thu cong).

23. TodoX API Contract V9 - API-first SaaS Foundation

Phan nay bo sung cac thong tin con thieu de chuyen TodoX tu n8n workflow sang he thong Web SaaS API-first. V8 tiep tục là lớp tương thích cho n8n, V9 là hướng chuẩn hóa lâu dài cho Dashboard, worker và client khác.

23.1 API Contract OpenAPI/Swagger

File contract chuẩn trong gói này: contracts/openapi-todox-v9.yaml. Khi cài đặt lên server nên copy vào:

E:\N8N.ANHDUC\todoX\contracts\openapi-todox-v9.yaml

Endpoint	Mục đích	Ghi chú
GET /health	Kiểm tra service	Không cần auth
POST /api/jobs	Tạo job TodoX	Dùng chung cho Telegram, API, Dashboard
POST /api/llm/director	LLM Gateway / AI Director	Router -> provider -> sdk -> model -> screenplay
GET /api/llm/providers	Xem router/provider/model đang active	Dùng để debug và Dashboard
POST /api/v8/main-prepare	V8 prepare từ n8n	Gửi tương thích
POST /api/v8/render-worker	V8 render worker batch	Gửi tương thích
POST /api/v8/finalize	V8 concat/final upload	Gửi tương thích
POST /api/jobs/worker-complete	Atomic worker counter	Đảm bảo concat chỉ chạy 1 lần
POST /api/storage/upload-json	Upload JSON artifact	Dùng relativePath, không dùng object_key

23.2 Database Schema chi tiết - LLM Gateway

Bảng	Vai trò
todox_llm_providers	Đăng ký provider/adaptor: CloudGemini, WokuGPT, WokuClaude, WokuGemini, WokuDeepSeek, OpenRouter...
todox_llm_models	Danh mục model theo provider, context, max tokens, capabilities và giá sau này
todox_llm_routers	Alias router từ Telegram/SaaS: GPT, Claude, Gemini, DeepSeek -> provider/model
todox_llm_requests	Log mọi request/response LLM, usage, latency, error, fallback
todox_worker_events	Event stream giữa các worker để Dashboard hiển thị theo thời gian thực

Column	todox_llm_providers
provider_key	Khoá ổn định, ví dụ CloudGemini, WokuGPT
provider_name	Tên hiển thị
sdk	vertex-gemini/openai-completions/anthropic-messages/google-generative-ai
base_url	Base URL của provider
api_key_setting	Key trong todox_settings chưa secret
enabled	Bật/tắt provider
options	JSON tùy biến

Router	Provider	SDK	Model	Billing
CloudGemini	Google Cloud	vertex-gemini	gemini-2.5-pro	Google Cloud
GPT	WokuGPT	openai-completions	gpt-4o	Woku
Claude	WokuClaude	anthropic-messages	claude-sonnet-4-6	Woku
DeepSeek	WokuDeepSeek	openai-completions	deepseek-chat	Woku
WokuGemini	WokuGemini	google-generative-ai	gemini-2.0-flash	Woku

23.3 LLM Gateway Adapter Rule

Không được chỉ map router theo tên model. Woku có nhiều adapter khác nhau nên phải lookup sdk từ database.

router -> todox_llm_routers -> provider/model -> todox_llm_providers.sdk -> adapter factory

SDK	Endpoint chuẩn	Ghi chú
openai-completions	{base_url}/chat/completions	Dùng cho WokuGPT, WokuDeepSeek, OpenRouter, OpenAI-compatible
anthropic-messages	{base_url}/v1/messages	Dùng cho WokuClaude/Anthropic-compatible
google-generative-ai	{base_url}/v1beta/models/{model}:generateContent	Dùng cho WokuGemini nếu provider hỗ trợ Google-compatible
vertex-gemini	Google Vertex generateContent	Dùng cho CloudGemini mặc định

23.4 Sequence Diagram - Render Pipeline

```
Telegram/User -> n8n Main: /render <url> 8 GPT
n8n Main -> API: POST /api/jobs
n8n Main -> API: POST /api/v8/main-prepare
API -> MinIO: upload metadata/analysis/remake
n8n Main -> API: POST /api/llm/director
API -> PostgreSQL: lookup router/provider/model + log todox_llm_requests
API -> Provider: call CloudGemini/Woku/OpenRouter...
API -> n8n Main: screenplay.scenePlan
n8n Main -> Worker1/2/3: fan-out webhooks, respond immediately
Workers -> API: POST /api/v8/render-worker
API -> MinIO: scene-video, scene-voice, scene-caption
Workers -> API: POST /api/jobs/worker-complete
Last worker -> Concat: webhook concat-deliver
Concat -> API: POST /api/v8/finalize
API -> MinIO: video-final/final_v1.mp4
Concat -> Telegram: final MinIO link
```

23.5 Sequence Diagram - AI Director Pipeline

```
n8n AI Director -> POST /api/llm/director
API -> PostgreSQL: load todox_llm_routers/providers/models/settings
API -> AdapterFactory: create adapter by sdk
Adapter -> Provider: call model
alt success
  Provider -> API: JSON screenplay
  API -> DB: todox_llm_requests status=success
  API -> n8n: fallback=false, screenplay
else provider error or invalid JSON
  API -> deterministic fallback
  API -> DB: todox_llm_requests status=fallback + fallbackReason
  API -> n8n: fallback=true, screenplay fallback
end
```

23.6 Sequence Diagram - Worker Pipeline

```
n8n Main -> n8n Worker: webhook payload
n8n Worker -> n8n Main: respond immediately
n8n Worker -> API: POST /api/v8/render-worker
loop each scene
  API -> Vertex/Veo: generate scene video
  API -> Google TTS: synthesize voice
  API -> FFmpeg: mux video + voice + caption
  API -> MinIO: upload scene artifacts
  API -> PostgreSQL: update todox_scene_render_tasks
end
n8n Worker -> API: POST /api/jobs/worker-complete
```

23.7 Error Code Standard

Code	HTTP	Y nghĩa	Hành động gợi ý
UNAUTHORIZED	401	Sai/thiếu X-TodoX-Secret	Kiểm tra api_secret trong SQL/n8n
BAD_REQUEST	400	Thiếu input bắt buộc	Kiểm tra body JSON
CONFIG_MISSING	400/500	Thiếu provider/model/key	Chạy check config
LLM_GATEWAY_FAILED	500	Gateway runtime error	Kiểm tra route/API log
LLM_PROVIDER_ERROR	200 fallback/502	Provider lỗi HTTP	Xem todox_llm_requests.error_json
LLM_INVALID_JSON	200 fallback	Model trả về không đúng schema	Tăng prompt/schema
LLM_FALLBACK_USED	200	Đã dùng deterministic fallback	Xem fallbackReason
MAIN_PREPARE_FAILED	500	Lỗi download/analysis/manifest	Xem job debug/log
WORKER_RENDER_FAILED	500	Worker lỗi toàn cục	Xem worker execution
SCENE_RENDER_FAILED	200 partial	Một scene lỗi	Xem todox_scene_render_tasks
FINALIZE_FAILED	500	Concat/upload final lỗi	Kiểm tra scene-video và MinIO
MINIO_UPLOAD_FAILED	500/pending	Upload MinIO lỗi	Kiểm tra mc alias/bucket

WEBHOOK_NOT_REGISTERED	404	n8n workflow chưa active/sai path	Active workflow và cập nhật SQL path
------------------------	-----	-----------------------------------	--------------------------------------

23.8 Versioning API

V8 là compatibility layer đang phục vụ n8n hiện tại. V9 là API-first cho SaaS. Không phá vỡ endpoint V8 khi xây V9.

Version	Path	Vai trò
V8	/api/v8/*	Compatibility cho workflow n8n hiện tại
V9	/api/v9/*	API-first cho SaaS Dashboard và worker mới
LLM Gateway	/api/llm/*	Dùng chung cho V8 và V9 để tránh duplicate logic

```

/api/v9/jobs
/api/v9/render/prepare
/api/v9/render/dispatch
/api/v9/render/finalize
/api/v9/llm/director
/api/v9/providers
/api/v9/models
/api/v9/routers
/api/v9/events

```

23.9 Authentication/Authorization

Layer	Co che	Ghi chú
Internal automation	X-TodoX-Secret	Dùng cho n8n/worker hiện tại
SaaS future	JWT access token	User đang nhập web
Workspace authorization	workspace_id/project_id	Mọi user chỉ thấy job của workspace mình
RBAC	owner/admin/operator/viewer	Phân quyền cấu hình provider và chạy job
Secret storage	encrypted at rest	API key Woku/OpenAI/Anthropic không lưu plain text lâu dài

23.10 Webhook Contract

Webhook	Method	Response mode	Payload chính
/webhook/worker-scene-1	POST	Respond immediately	runtime.jobId, workerId, scenes, CONFIG
/webhook/worker-scene-2	POST	Respond immediately	runtime.jobId, workerId, scenes, CONFIG
/webhook/worker-scene-3	POST	Respond immediately	runtime.jobId, workerId, scenes, CONFIG
/webhook/concat-deliver	POST	lastNode OK	runtime.jobId, chatId, CONFIG
/webhook/todoxx-v8-ai-director	POST	lastNode/Immediate tùy thiết kế	Nên thay bằng /api/llm/director lâu dài

```

{
  "runtime": {
    "jobId": "job_xxx",
    "workerId": "1",
    "chatId": "5571707300",
    "sceneCount": 8,
    "workerTotal": 3,
    "scenes": [],
    "CONFIG": {}
  }
}

```

23.11 Event Contract between workers

Event nên là append-only. Dashboard Web doc todoxx_worker_events + todoxx_jobs + todoxx_scene_render_tasks để hiển thị theo thời gian thực.

```

{
  "eventId": "evt_xxx",
  "jobId": "job_xxx",
  "workerId": "worker-1",

```



```

"eventType": "worker.scene.completed",
"status": "completed",
"payload": {},
"createdAt": "2026-06-29T00:00:00Z"
}

```

Event	Producer	Purpose
job.created	API	Job created
source.download.started	API	Source download started
source.download.completed	API	Source ready
scene.analysis.completed	API	Raw visual analysis complete
llm.director.started	API	LLM screenplay started
llm.director.completed	API	LLM screenplay complete
llm.director.fallback	API	Fallback was used
worker.started	Worker	Worker accepted batch
worker.scene.started	Worker/API	Scene started
worker.scene.completed	Worker/API	Scene completed
worker.scene.failed	Worker/API	Scene failed
worker.completed	Worker/API	Batch completed
finalize.started	Concat/API	Final concat started
finalize.completed	API	Final video ready
delivery.telegram.sent	n8n/API	Telegram sent

23.12 Nội dung tài liệu cu đang thiếu

- Chưa có LLM Gateway adapter factory theo SDK. Đây là nguyên nhân để router GPT có thể bị gọi nhầm Claude distributor.
- Chưa tách CloudGemini và WokuGemini. Cần giữ CloudGemini làm mặc định vì khác lương phí.
- Chưa có bang todox_llm_requests để biết rằng đã gọi Woku hay fallback.
- Chưa có debug file llm_request/llm_response trong job folder.
- Chưa có error code chuẩn cho LLM_PROVIDER_ERROR, LLM_INVALID_JSON, LLM_FALLBACK_USED.
- Chưa có event contract cho dashboard SaaS realtime.
- Chưa có versioning /api/v8 và /api/v9 rõ ràng.
- Chưa có authorization cho Web SaaS: JWT, workspace, RBAC, encrypted provider secrets.

26. TodoX V9 Database User Standard

Mục này bổ sung chuẩn phân tách database/user cho TodoX V9. n8n tiếp tục sử dụng database n8n_db với user n8n_user. TodoX Automation API sử dụng database todox với user ứng dụng riêng todox_user thay vì dùng superuser postgres trong runtime.

26.1 Current Separation

System	Database	User	Purpose
n8n	n8n_db	n8n_user	Lưu workflow, credentials và execution của n8n.
TodoX	todox	todox_user	Lưu job, scene, LLM, agent, worker, artifact metadata của TodoX.
Admin	postgres	postgres	Chỉ dùng để migration/admin, không dùng cho runtime ứng dụng.

26.2 Permission Model

- todox_user được GRANT CONNECT trên database todox.
- todox_user được GRANT USAGE trên schema public.
- todox_user được SELECT, INSERT, UPDATE, DELETE trên tất cả bảng public.
- todox_user được USAGE, SELECT, UPDATE trên sequence.
- todox_user được EXECUTE trên function.
- Default privileges được set để bảng/sequence/function mới cung cấp quyền cho todox_user.
- Không cấp CREATEDB, CREATEROLE hoặc SUPERUSER cho todox_user.

26.3 Runtime Environment Variables

Variable	Recommended Value
TODOX_PGHOST	127.0.0.1
TODOX_PGPORT	5432
TODOX_PGDATABASE	todox
TODOX_PGUSER	todox_user
TODOX_PGPASSWORD	toDoX@123#2026

26.4 Install/Update Scripts

- 01_Create-TodoX-DbUser.ps1: tạo/cấp nhật role todox_user và grant quyền trên database todox.
- 02_Check-TodoX-DbUser.ps1: kiểm tra todox_user kết nối và đọc được bảng TodoX.
- 03_Update-TodoXAutomationAPI-DbEnv.ps1: cấp nhật NSSM AppEnvironmentExtra cho TodoXAutomationAPI, có backup trước khi sửa.
- 04_List-Postgres-Databases-And-Users.ps1: liệt kê database/user/owner hiện có trên PostgreSQL.

26.5 Operational Rule

Sau V9, mọi script runtime của TodoX nên ưu tiên dùng todox_user. Tài khoản postgres chỉ dùng cho migration, backup, restore, grant quyền và tác vụ quản trị. Không thay đổi n8n_db/n8n_user khi cấp nhật TodoX.